

Hypothesis Testing

The scientific method in action



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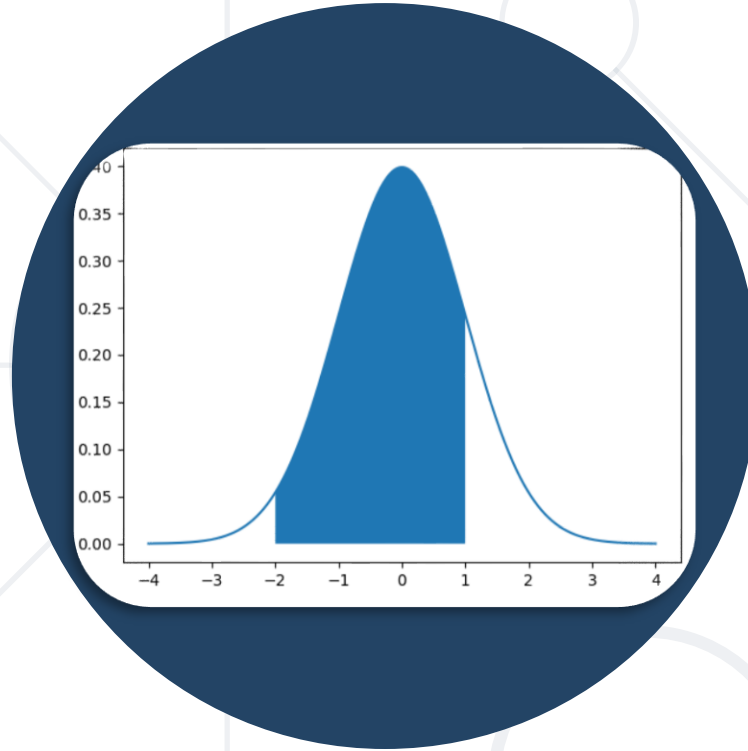


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- Confidence intervals
 - Confidence level
- Hypothesis tests
 - Z-test
 - t-test (one-sample, two-sample)
- Hypothesis tests of many variables
 - ANOVA
 - Chi-squared
- p-value misconceptions





Confidence Intervals

Being confident is important

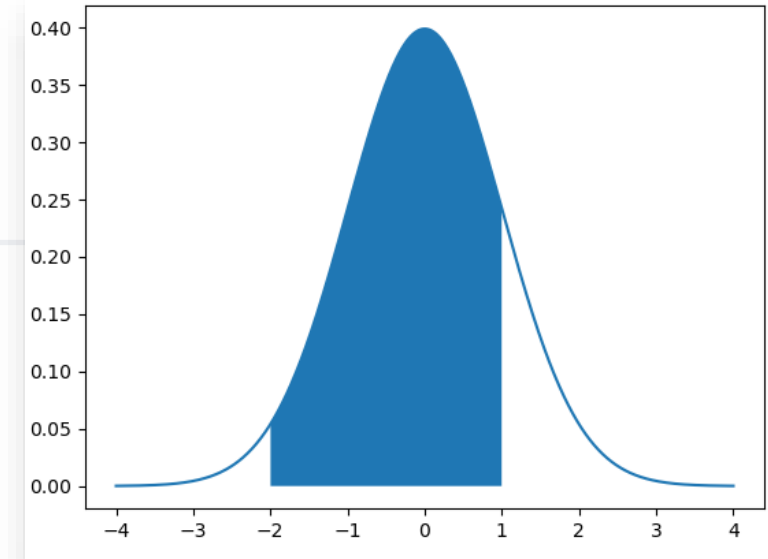
- In an experiment, we can't observe the variables' true values directly
 - We observe other values
 - We make assumptions as to how they are distributed
 - We can estimate the true value
 - **Law of large numbers:** when our sample is big enough, the sample parameters approach the population parameters
- With continuous values, it's useless to say that the mean is equal to a certain value (why?)
- **Confidence interval** – a range of values that we're fairly sure contains the true value
 - **How confident?** A matter of choice
- **Confidence level** – the probability that the value falls within the interval

Confidence Intervals – Interpretation

- Similar to the probability interpretations
- To illustrate these, let's take a confidence interval $[5; 7,3]$ and a **70%** confidence level
- Frequency
 - If we perform the experiment many times, 70% of the values will fall in the interval $[5; 7,3]$ and 30% – outside it
- Certainty of next trial
 - Next time we perform the experiment, we are 70% certain that the value will fall within $[5; 7,3]$
 - Note that this is a statement **about the interval**, not about the value
- Typically used confidence levels
 - **50%; 90%; 95%; 99,7%**

Confidence Intervals and Z-Scores

- Observe the Z-distribution (Gaussian, $\mu = 0$, $\sigma = 1$)
- What's the probability that a value drawn from it $x \in [-2; 1]$?
 - This corresponds to the shaded area in the graph
 - The cumulative function gives us the area to the left of some value
 - Shaded area = $cdf(1) - cdf(-2) = 0,819 = 81,9\%$
- Interpretations
 - If we draw many random numbers from the Z-distribution, we expect that 81,9% of them will be in $[-2; 1]$
 - If we draw one random number, there is 81,9% chance of it being in $[-2; 1]$
- Commonly used intervals
 - $1\sigma \rightarrow 68,27\%$; $2\sigma \rightarrow 95,45\%$; $3\sigma \rightarrow 99,73\%$
 - Also $1,96\sigma \rightarrow 95\%$



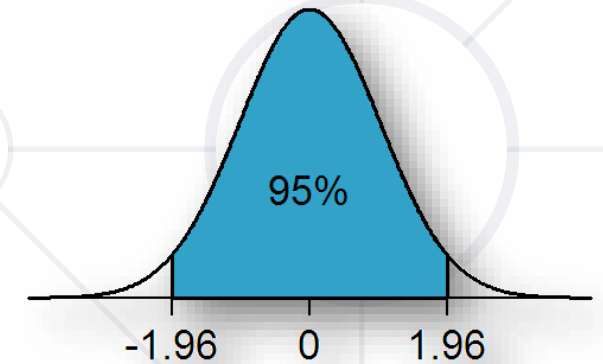
Confidence Intervals: Example

- In the dataset heights.dat you're given the measured heights (in cm) of 351 elderly women (from an osteoporosis study)
 - Plot a histogram and / or boxplot to see what the distribution is
 - Print the mean $\bar{x}$ and standard deviation s of the sample
 - Assume that the population follows a normal distribution
 - Real parameters – unknown; our best guess: $\mu = \bar{x}, \sigma = s$
 - What are the confidence intervals of
 - 50%, 90%, 95%

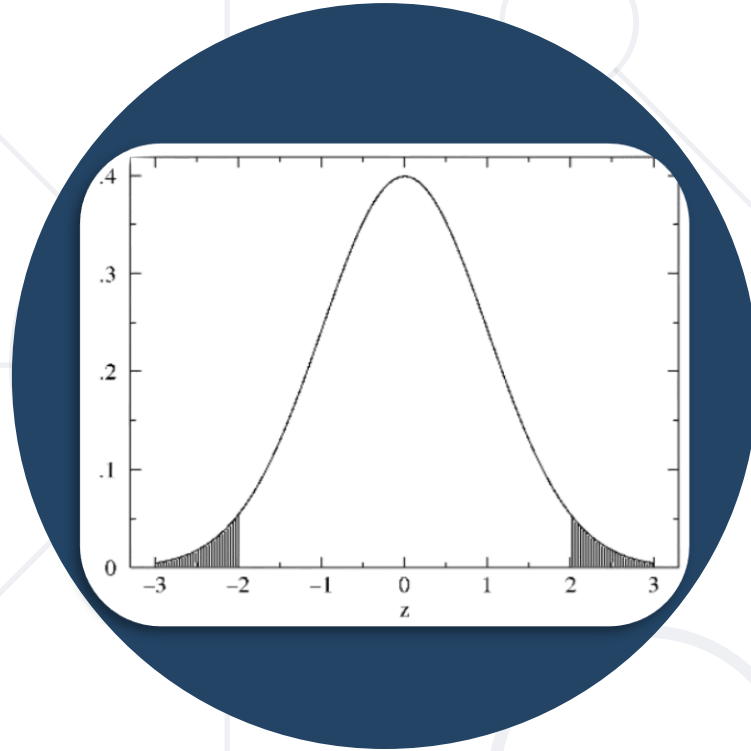
- To calculate the confidence intervals, we need to calculate the **Z-scores**
 - To do this, we'll use the percent point function, ppf
 - Inverse of the cdf
 - Returns the value at which the probability is less than or equal to the given probability
 - Example: Z-distribution
 - $ppf(0) = -\infty$; $ppf(1) = \infty$; $ppf(0,5) = 0$; $ppf(0,975) = 1,96$

Confidence Intervals Example

- Note that once again we need to subtract the left white region
 - Area of shaded region: p (e.g., $p = 0,95$)
 - Area of both tails: $1 - p$
 - Percentage point of left tail: $\frac{1-p}{2}$
 - Percentage point of right tail: $\frac{1-p}{2} + p = \frac{1-p+2p}{2} = \frac{1+p}{2}$



```
import scipy.stats as st
def get_real_confidence_interval(probability, mean, std):
    lower_area = (1 - probability) / 2
    upper_area = (1 + probability) / 2
    return [
        st.norm.ppf(lower_area, mean, std),
        st.norm.ppf(upper_area, mean, std)]
```



Testing Hypotheses

The scientific method in action

- After performing an experiment and getting data, the scientific method requires that we form a hypothesis
 - Fact, law, theory and hypothesis are different terms
- In the simplest case, we have two hypotheses
 - **Null hypothesis** (H_0) – the status quo is real, "nothing interesting happens"
 - **Alternate hypothesis** (H_1) – what we're trying to demonstrate
- Types of hypotheses
 - **Attributive** – something exists and can be measured
 - **Associative** – there is a relationship between two behaviors
 - **Causal** – differences in the amount / kind of one behavior cause differences in other behaviors

- Examples of hypotheses – study of Disneyland visitors
 - **Attributive**
 - Most of the population has heard of Disneyland
 - Disneyland visitors are diverse in demographics
 - **Associative**
 - Income level is correlated with visiting Disneyland
 - People who live closer to Disneyland are more apt to visit Disneyland
 - **Causal**
 - Frequent exposure to Disneyland advertising results in increased attendance
 - Discounting tickets for local residents produces an increase in visitor numbers
- Note that attributive hypotheses involve one variable (univariate) while associative and causal hypotheses involve two variables (bivariate)

Testing a Hypothesis

- In random experiments, we have error sources
 - Human error, systematic error, random errors, etc.
- We cannot prove (or reject) a hypothesis with complete certainty
- The errors we can make are two types
 - **Type I error** – reject H_0 while it's true (false positive)
 - **Type II error** – accept H_0 while H_1 is true (false negative)
- The possible results can be summarized in the following truth table
 - Also called **confusion matrix**

		Action	
		Don't reject H_0	Reject H_0
Reality	H_0 true	TN true negative	FP (type I error) false positive
	H_0 false	FN (type II error) false negative	TP true positive

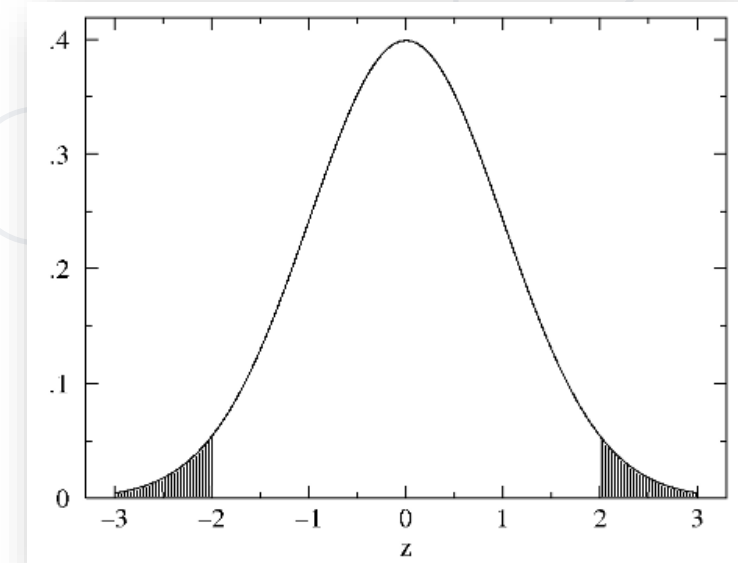
- To measure the probability of producing a wrong hypothesis, we use a **test statistic** – measure of deviations from H_0
 - Different tests produce different measures (statistics)
 - **We accept or reject the null hypothesis based on the value of the test statistic**
- Let's denote **the probability of getting a type I error** with α
 - Each value of the selected test statistic has a corresponding alpha-value
 - We perform the experiment, get data and calculate the test statistic value
 - From that, we calculate the corresponding alpha-value
 - We reject the null hypothesis if $\alpha < \alpha_c$, where α_c is a **critical confidence level**

- A Z-test uses the Z-statistic
- H_0 : standard normal distribution
- Example: light bulb factory
 - A factory produces light bulbs with lifetime $X \sim N(\mu = 500h, \sigma = 50h)$
 - A sample of 25 bulbs has a mean lifetime $\bar{x} = 480h$
 - Is there something wrong with the production line?
- Forming hypotheses
 - H_0 : The production line works normally; the observed deviation of the sample mean from the population mean is due to chance
 - H_1 : The production line is broken

- Suppose we take a lot of samples from the entire population
 - Each sample mean will be different
 - The distribution of sample means will be more or less Gaussian
 - Parameters (our best estimate): $\mu_{\bar{x}} = \mu, \sigma_{\bar{x}} = \sigma/\sqrt{n}$
 - Here's why the parameters are chosen like this
- If H_0 is correct, we assume that $\bar{x} \sim N(\mu, \sigma/\sqrt{n})$
- Z-statistic
 - $Z = \frac{\bar{x} - \mu}{\sigma_{\bar{x}}} = \frac{480 - 500}{50/\sqrt{25}} = -2$
- We can see that we are 2 std's below the mean
- How extreme is that?
 - What's the probability that we get results **as extreme or more extreme** than we observed, assuming the null hypothesis is true?
 - Less than 5%

Two-tailed Z-test

- We can get the confidence interval from the Z-statistic
- We are looking for **more extreme** values
 - Values **outside** the confidence interval
 - What's the probability $P(|Z| \geq 2)$?
 - We're looking for a value different than the mean
 - We **can't assume** whether it's smaller or larger
 - Therefore, we have to look at both "tails" of the distribution
- If we assume a critical value (also called a p-value) of 5%, **the results are significant**
 - $P(|Z| > 2) \approx 0,0455 = 4,55\%$
- We can **reject H_0 at the 5% level**
 - Even at lower levels, up to 4,55%



- The same logic applies, but now we're looking at one tail only
- Question: Is the lifespan **significantly lower** than it should be?
Cutoff point: $\alpha_c = 5\%$, $Z = -2$
 - $P(Z \leq -2) = \frac{0,00455}{2} - 0,02275 = 2,275\% < \alpha_c$
 - Answer: Yes, at the given significance level
- Question: Is the lifespan **significantly higher** than it should be?
 - $P(Z \geq -2) = 97,725\% \gg \alpha_c$
 - Answer: No, at the given significance level

- The Z-test requires that we know the standard deviation of the population
 - Usually not available
- We can use another test statistic, called **t**
- Advantages over the Z-test
 - We don't need to know the population σ
 - It's better when we have very small sample sizes (e.g., $n < 30$)
 - It can be used for testing the mean of a sample against a standard, but also, for comparing two means
 - We can see whether two sets of data are significantly different from each other
- **Null hypothesis:** The test statistic follows Student's t-distribution
 - Similar to Gaussian distribution, with "fatter" tails

- The details of the calculation are fairly complex but we can do this in code
 - Using `scipy.stats`
- First, we generate 100 random numbers with $\mu = 5, \sigma = 10$
- Then we ask whether the sample mean is equal to the true mean (and other values, just for testing)
- We get the p-value – probability of the null hypothesis being true
 - I.e., probability that the mean is equal to the given mean

```
sample_data = st.norm.rvs(5, 10, 100)

print(st.ttest_1samp(sample_data, 5).pvalue) # 0.9301
print(st.ttest_1samp(sample_data, 4).pvalue) # 0.3352
print(st.ttest_1samp(sample_data, 0).pvalue) # 1.104e-6
```

Independent Two-Sample t-test

- We compare two independent distributions
 - We want to see whether they have the same mean
 - We assume equal variances (scipy can also do tests with unequal variances – important when sample sizes differ)
- Example: Grain size
 - We are given data (in grain_data.dat) of grain sizes from two different farms
 - Do they differ significantly (at the 95% level)?
 - * We can also plot histograms to see what the distributions look like

```
grain_data = ...  
st.ttest_ind(grain_data.GreatNorthern, grain_data.BigFour)  
# Ttest_indResult(statistic=1.312336706487564,  
# pvalue=0.20792200785311768)
```

Paired Two-Sample t-test

- We compare two distributions
 - Observations in samples can be paired
 - Examples – before / after observations; comparison between two different treatments applied to the same subjects
- Example: Drinking water
 - We are given data (in water_data.dat) of Zn concentration in surface and bottom water at 10 different locations
 - Does the true average concentration in bottom water exceed that of top water?
 - We use a paired t-test because the samples are from the same locations
 - It reduces experimental error (and provides stronger evidence)

```
water_data = ...  
# We use a one-tailed t-test  
st.ttest_rel(water_data.surface, water_data.bottom).pvalue / 2  
# 0.00044555772891127738
```

- Sometimes it's not enough to compare two distributions
 - We may want to compare multiple distributions against the same null hypothesis
 - E.g., how is the percentage of smokers distributed by income and age?
- Other times, we create a model and want to evaluate it
 - E.g., a linear regression
 - We can explain some of the variance in the sample
- There are other tests to perform these "checks"
 - **ANOVA** (Analysis of Variance) – useful for grouped data
 - Observe the variance inside groups and between groups
 - **Chi-square(d) test** – can be applied to categorical data
 - Two common types
 - How good a model is (goodness of fit)
 - Whether two variables are independent

Analysis of Variance (ANOVA)

- We want to compare several **groups**
- H_0 : The means of the groups are the same
- Method (*[scipy.stats.f oneway\(\)](#)*)
 - For each group $\Rightarrow$ group mean
 - In-group variance: distances from an individual point to the group mean
 - Between-group variance: distances between the means of two groups
 - For the entire data $\Rightarrow$ total mean (mean of all data)
 - Also equal to the mean of all group means
 - Total variance: in-group + between-group
- F-statistic (Fisher)
 - $F = \frac{\text{variance between groups}}{\text{variance within groups}}$
 - F – large $\Rightarrow$ the variance between groups dominates
 - For each value of F , there's a corresponding p -value
 - If $p \leq p_c$, we can reject H_0

Chi-Squared (χ^2) Test

- Compares expected (predicted) and observed frequencies
 - Is there a significant difference between these?
 - Used to compare **categories** (one against another)
 - Compare to ANOVA – numbers w.r.t. categories
 - May also be used as a goodness-of-fit measure
 - How well were we able to predict
- Statistic: $\chi^2 = \frac{(f_{\text{observed}} - f_{\text{estimated}})^2}{f_{\text{estimated}}}$
- H_0 : No significant difference between observed and estimated frequencies among the categories (groups)
 - The test returns the value of the statistic and the p-value corresponding to it
 - Works the same as any other test
 - Python: [scipy.stats.chisquare\(\)](#)



Common Misconceptions

Because everyone can be wrong

Some p-value Misconceptions

- Goodman, S. (2008), [source](#)
- "If $p = 0,05$, H_0 has 5% chance of being true"
 - **The data alone can't tell us how likely we are to be wrong**
 - p is calculated under H_0 , so it can't be the probability of H_0 being false
- " $p = 0,05$ means that if we reject H_0 , the probability of type I error (false positive) is only 5%"
 - I.e., seeing a difference where there isn't any
 - $\Rightarrow$ 5% chance of false rejection = 5% chance H_0 is true
 - Wrong, see first bullet
- "If $p = 0,05$, we have observed data that will occur **only** 5% of the time assuming H_0 "
 - The p-value is the probability of observing data **as extreme or more extreme** under H_0

Some p-value Misconceptions

- "A nonsignificant difference means the groups are the same"
 - It only means **we don't have enough data** to reject H_0
- "A scientific conclusion or treatment policy must be based on whether or not the p -value is significant"
 - **The results have to be checked** against prior data
- Failing to reject H_0 means that H_0 is true
 - It means that we don't have enough evidence to reject it
 - **We can't accept (or reject) any other hypothesis**
 - *"Absence of evidence is not evidence of absence"*
- <https://xkcd.com/882/>
- <https://www.xkcd.com/1478/>
- ["Still. Not. Significant"](#) article

Summary

- Confidence intervals
 - Confidence level
- Hypothesis tests
 - Z-test
 - t-test (one-sample, two-sample)
- Hypothesis tests of many variables
 - ANOVA
 - Chi-squared
- p-value misconceptions



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